

8(a)	region (of space) where there is a force	M1
	produced by / on a magnet / magnetic pole / <u>moving</u> charge / current-carrying conductor	A1
8(b)(i)	out of (the plane of) the paper / page	B1
8(b)(ii)	the force on the particle is (always) perpendicular to the velocity / perpendicular to the direction of travel / towards the centre of path	B1
	no work is done by the force on the particle / there is no acceleration in the direction of the velocity / the acceleration is (always) perpendicular to the velocity	B1
8(b)(iii)	$F = Bqv$ or $F = mv^2/r$	C1
	$mv^2/(d/2) = Bqv$ so $d = 2mv/Bq$	A1
8(b)(iv)	time = distance / speed $T_{(F)} = \pi d / 2v$	C1
	$T_{(F)} = (\pi / 2v) \times (2mv / Bq)$ $T_{(F)} = \pi m / Bq$ and so $T_{(F)}$ independent of v	A1